

Composing an Authority Layer with a Sandbox Runtime: BULL on NVIDIA OpenShell

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Abstract—A sandbox runtime decides *where* an agent’s code may reach; an authority layer decides *whether* a specific action, *given where its inputs came from, is authorized and by whom*. This report tests whether the two can be composed without the composition itself opening a bypass. BULL, an execution-governance runtime with signed grants, provenance tracking, single-use human approvals, and a hash-chained ledger, was implemented as an NVIDIA OpenShell supervisor middleware and gateway interceptor and evaluated against a live OpenShell v0.1.2 gateway built from source (commit 6648bd0) on the Docker compute driver. The invariant under test is $\text{Effect}(r) \Rightarrow \text{BULL}(r) \wedge \text{OpenShell}(r)$, with absence of BULL implying no effect. A ten-case proof suite plus two added cases was run in two phases—OpenShell alone and OpenShell with BULL registered fail-closed—and every outcome was judged at the destination service, not from client output. In the final run 26 of 26 checks behaved as specified; with BULL, 14 of 14 properties held, including approval-gated exfiltration after untrusted input, refusal of unapproved policy widening and credential attachment, fail-closed behavior when BULL was killed, and a join of 9 of 9 BULL decisions to OpenShell’s OCSF telemetry with zero effects lacking a BULL allow. Composition added a median 8.08 ms per GET (95% CI [7.12, 8.86]) and 10.98 ms per POST ([10.16, 11.72]), $n = 200$ each. The experiment also surfaced findings about both systems: OpenShell’s L7 rules default to an audit mode that forwards violations, its middleware requests misreport scheme and omit the originating process, and BULL’s ledger append is $O(n)$ in ledger length. Six integration defects in BULL were found only against the live gateway and fixed. These are project-authored results from a single host, not independent certification.

Index Terms—AI-agent security, execution governance, sandbox runtime, OpenShell, supervisor middleware, gateway interceptor, provenance, fail-closed, OCSF, audit integrity.

I. INTRODUCTION

AGENT sandboxes and agent authority layers answer different questions. NVIDIA OpenShell runs each agent in a sandbox whose declarative network and filesystem policy, enforced by a per-sandbox supervisor, bounds the hosts, ports, methods, paths, and binaries the workload can use, and it emits OCSF security telemetry for what it allows and denies [1], [7]. It does not know that the text an agent just read came from an attacker, and anyone able to call its configuration API can widen a policy. BULL treats agent output as a proposal and binds execution to host-owned authority, provenance, and approvals [15]; but a decision layer binds only the traffic that passes through it.

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OpenShell exposes two extension points that make composition possible without modifying either system: a *supervisor middleware* called for each proxied HTTP request after OpenShell’s own policy and before credential injection, and a *gateway interceptor* called on control-plane RPCs, able to patch a proposed operation or refuse it [2], [3]. This report asks three questions. **RQ1:** Can BULL be composed through these extension points as a narrowing-only layer, such that an effect requires both systems’ consent? **RQ2:** Does the composition fail closed when BULL is absent, and can the two systems’ records be joined into one trail? **RQ3:** What does the composition cost, and what does exercising it reveal about either component?

A. Scope and Evidence Discipline

This report separates four evidence classes: (1) upstream OpenShell source and documentation at tag `v0.1.2`; (2) BULL source on the experiment branch; (3) measurements from one automated two-phase run on one host, whose raw files are hashed in Supplement S6; and (4) limitations and experiments not performed. All tables, figures, and in-text numbers are generated from the evidence files by the build script; none are transcribed by hand. A passing case demonstrates the stated behavior for this configuration; it is not proof against unknown attacks.

II. THREAT MODEL AND COMPOSITION CONTRACT

A. Adversaries and Assets

The untrusted side is the sandboxed workload and everything it reads: model output, fetched content, and processes it starts. Three adversary behaviors are exercised: a workload that attempts an effect it was not granted; a workload steered by untrusted content toward exfiltration through an allowed endpoint; and a caller of the OpenShell API who attempts to widen a sandbox’s authority or attach a credential. Protected assets are effects at destination services, provider credentials, and the integrity of the combined audit record.

B. Trusted Computing Base

Both systems are trusted: the OpenShell gateway, supervisor, and sandbox runtime; the Docker daemon and Linux kernel; the BULL authority process, its signing key, and its ledger; and the operator who approves escalations. A malicious host administrator and a compromised kernel or container runtime are out of scope. In this experiment the middleware transport was not authenticated (Section IX); that is a deviation from a production trust configuration, not an assumption about it.

C. Contract

Let r be a request from sandbox s . Let $\text{OS}(r)$ denote OpenShell’s network and L7 policy decision and $\text{B}(r)$ BULL’s decision. The composition is intended to satisfy

$$\begin{aligned} \text{Effect}(r) &\implies \text{OS}(r) \wedge \text{B}(r), \\ \neg \text{Available}(\text{BULL}) &\implies \neg \text{Effect}(r) \wedge \neg \text{Admit}(s'), \\ \text{Policy}_{t+1}(s) \supsetneq \text{Policy}_t(s) &\implies \text{Approved}(\Delta), \\ \text{Effect}(r) &\implies \exists \ell \in \text{Ledger} : \ell \sim r, \end{aligned}$$

where s' is a sandbox being created, Δ the added endpoints, and $\ell \sim r$ a ledger record joinable to r ’s OCSF events. B is *narrowing only*: the middleware can return allow or deny, never widen OpenShell’s decision, and the interceptor can only attach BULL’s middleware or refuse.

III. DESIGN

A. Data Plane: Supervisor Middleware

BULL registers one binding: HTTP request, phase `PRE_CREDENTIALS`, 256 KiB payload ceiling [4]. For each request, `GET/HEAD/OPTIONS` map to `capability.network.outbound` and other methods to `network.post`; the sandbox’s signed grant must contain the capability. A sandbox’s provenance becomes `internet` once it performs an allowed read from a `host:port` not listed as trusted; afterwards BULL’s deterministic policy escalates any `network.post`. An escalation creates a pending approval bound to method, URL, and body hash; an operator’s approval admits that exact request once. Every decision is appended to BULL’s hash-chained ledger. Taint and approvals persist across restarts.

B. Control Plane: Gateway Interceptor

BULL governs `CreateSandbox` and `UpdateConfig` in the `modify_operation` and `validate` phases, and `AttachSandboxProvider`, `ApproveDraftChunk`, `ApproveAllDraftChunks`, and `EditDraftChunk` in `validate` [5]. `modify_operation` returns RFC 6902 patches [13] that attach BULL’s middleware with `onError: fail_closed` to every host of a new or replacement policy. `validate` refuses: sandboxes not fully mediated; hosts outside the signed ceiling; L7 rules not in `enforce` mode; any new `host:port` or new L7 rule set without a matching single-use approval; all draft-policy approvals without BULL approval; global policy changes; and provider attachment for sandboxes lacking a `credential.read` grant. Both extensions are registered with `failure_policy = fail_closed`.

C. Grants and Audit Join

Per-sandbox grants (capabilities, trusted `host:port` sources, and a global host ceiling) are embedded in BULL’s HMAC-signed policy bundle. For the audit join, OpenShell’s supervisor pushes OCSF events to the gateway, which serves them in shorthand [7]. A correlator matches each BULL ledger decision to the OpenShell L7 and middleware events of

TABLE I
EXPERIMENTAL ENVIRONMENT

Item	Value
OpenShell	v0.1.2, tag v0.1.2, commit 6648bd0; gateway, CLI, supervisor glibc; sandbox runtime static musl; Rust 1.95.0
Compute driver	Docker version 29.3.1, build c2be9cc, pull policy <code>never</code>
Host	Linux 6.18.44-fc-v37, 4 vCPU, 15 GiB, shared cloud container
Sandbox image BULL	<code>ubuntu:24.04 + curl, python3; uid 1500</code> commit 56faa30; Python 3.11; grpcio 1.84; protobuf 7.36
Transport	middleware: plaintext gRPC on bridge; interceptor: Unix socket

the same sandbox, method, and target within a time window, and assigns destination receipts only to decisions BULL allowed; any receipt left unassigned is an effect BULL never allowed.

IV. METHOD

A. Environment and Deviations

Table I records the environment. The session’s network policy blocked the official installer and image registries, so all binaries and images were built locally from the upstream source and Dockerfiles, with two deviations: the supervisor image used `distroless/cc-debian13` instead of the pinned `base-nossl-debian13`, because a locally linked supervisor requires `libgcc_s`; and `host.openshell.internal` was mapped on the host to the Docker bridge (172.17.0.1) so that the host-resident gateway reaches the middleware at the address the supervisors use. Policies list `allowed_ips: [172.17.0.1/32]` because OpenShell’s SSRF guard otherwise refuses private destinations [7].

B. Topology and Grants

Three host services record every request that reaches them: *docs* (trusted by BULL), *news* (untrusted; returns a prompt-injection string), and *api* (the effect target). OpenShell’s policy allows `GET /**` on *docs* and *news* and `POST /v1/**` on *api*, each with `enforcement: enforce` and binary `/usr/bin/curl`. BULL grants sandbox *coder* outbound and *post*, *reader* outbound only, and *bench* outbound and *post*; all trust only *docs*; the ceiling is `host.openshell.internal`.

C. Phases, Judging, and Measurement

Phase A runs OpenShell alone; Phase B runs the same requests with both BULL extensions registered. A case passes on destination evidence: a request is “blocked” only if the destination never received it. Phase A rows for BULL-specific properties record what OpenShell alone does; their “as expected” status means the observation was made, not that the property holds. Latency is `curl’s time_total` for

200 trusted GETs and 200 allowed POSTs in one in-sandbox loop, excluding exec overhead; admission time is `sandbox create -detach` wall time over 6 creations and measures admission including interceptor round trips, not boot to Ready. The harness (`tools/openshell_experiment.py`) is fully automated.

V. RESULTS

A. Proof Suite

Table II and Fig. 1 give the outcomes. In Phase A, 12 of 12 observations were as expected; in Phase B, 14 of 14 properties held (both counts include two setup rows). The complete observed output of every case is reproduced in Supplement S1.

Three results carry most of the weight. First, in C2, C4, and C7, OpenShell’s own L7 engine allowed every request (Table III), so the blocked effects in Phase B are attributable to BULL alone: without an authority layer, a sandbox that read attacker-controlled content posted to the allowed API (C7-A). Second, C4 shows exactly-once semantics: a pending escalation produced no effect, an approval admitted one request, and reusing the approval was refused, leaving exactly one upstream record. Third, C9 killed the BULL process: the next POST was denied by OpenShell with `middleware_failed: external_service_error` and never reached the destination, sandbox creation was refused, and after restart a trusted read succeeded with persisted taint and approvals intact.

B. Unified Audit Trail

Table III lists every Phase B BULL decision joined to OpenShell’s telemetry and to the destination. All 9 of 9 decisions matched an OpenShell middleware event, 9 were consistent in decision and effect, and 0 destination receipts lacked a BULL allow. The single OpenShell middleware event with no BULL decision (1) is the C9 fail-closed denial issued while BULL was down. The ledger verified intact at 19 records at the time of C10. OpenShell carries BULL’s reason code into its own denial telemetry (e.g., `middleware_denied:bull-governance: bull_approval_required`), so an OCSF-only consumer still sees why BULL refused.

C. Latency

Table IV and Figs. 2–5 summarize performance. Composition increased median request latency from 8.98 to 17.07 ms for GET and from 9.85 to 20.83 ms for POST. Table V gives bootstrap confidence intervals (10000 resamples, fixed seed) and two-sided Mann–Whitney tests. From OCSF timestamps, OpenShell’s L7 event preceded BULL’s verdict by a median of 5 ms (range 1–7 ms, $n = 10$, 1 ms resolution); this interval contains the plaintext gRPC round trip, the Python decision, and a synchronous ledger append. Mean admission time rose by 71 ms ($n = 6$), which is within run-to-run spread.

Phase B latency drifts upward within each loop (Fig. 4, Table VI): the GET median moved from 14.89 to 17.67 ms and the POST median from 19.49 to 25.13 ms, while Phase A stayed near constant. BULL’s AuditLedger re-verifies the

entire hash chain before each append, making append cost linear in ledger length; the ledger held 431 records at the end of Phase B. This is the most plausible cause but was not isolated experimentally. It also explains why POST overhead exceeds GET overhead: the POST loop ran second against a longer ledger.

VI. FINDINGS ABOUT THE COMPONENTS

F1: OpenShell L7 rules default to audit. An endpoint with L7 rules and no `enforcement` field logs violations and forwards them; OpenShell documents audit as the default [6]. In C3b-A, a POST outside the only allowed path reached the destination, logged as `ALLOWED ... engine:l7`. BULL now refuses such policies at creation, replacement, and rule merge, accepting both the YAML form and the protobuf-JSON enum name `NETWORK_ENFORCEMENT_MODE_ENFORCE` [14]; `UNSPECIFIED` is treated as audit.

F2: Transparent interception held. Policy DNS resolved the destination to a trap address, so `curl -noproxy` traffic was still captured, L7-evaluated, and in Phase B sent through BULL, which escalated it. Binary identity refused a Python client to an endpoint `curl` was allowed to use; a raw socket to a public address and UDP DNS both failed (C8).

F3: Middleware metadata gaps in v0.1.2. On the relay paths exercised, OpenShell passes the literal scheme `https` to middleware for plaintext HTTP while its own OCSF events report `http` [10], and `RequestContext.originating_process` is set to `None` for operator middleware [9], so BULL cannot bind a decision to the executable. BULL therefore does not trust scheme; the correlator compares host, port, and path only.

F4: The extension contract is strict, and its failures closed. A middleware reply whose `reason_code` does not match `[a-z][a-z0-9_]*` is treated as a middleware failure and the request is denied [9]; an extension not declaring `openshell.<family>.contract` is refused at gateway start [11]. Both mistakes appeared during integration and both stopped traffic rather than admitting it.

F5: OCSF JSONL export requires a writable `/var/log` in the supervisor container [8], which the Docker driver’s image did not provide; the same events were obtained through the gateway’s log API.

F6: `openshell sandbox exec` reads a non-TTY stdin to EOF before dispatching [12]; automation inheriting an open stdin blocks indefinitely.

VII. DEFECTS FOUND IN BULL

Six defects passed BULL’s unit tests and surfaced only against the live gateway (Table VII). Each has a regression test in `tests/test_openshell_authority.py`. Defect 3 is the most consequential: endpoints rendered as `host:40605.0` would never equal later `host:40605` comparisons, a latent source of false widening or missed widening.

TABLE II
PROOF-SUITE OUTCOMES, JUDGED AT THE DESTINATION

Case	Property	A: OpenShell only	B: OpenShell + BULL	B
C1	BULL ALLOW $\wedge$ OpenShell ALLOW executes	HTTP 200; effect reached upstream	HTTP 200; effect reached upstream	held
C2	BULL DENY $\wedge$ OpenShell ALLOW is blocked	HTTP 200; effect reached upstream	HTTP 403; no effect	held
C3	BULL ALLOW $\wedge$ OpenShell DENY is blocked	HTTP 403; no effect	HTTP 403; no effect	held
C3b	L7 rules left at default (audit) enforcement	created; out-of-rule POST reached upstream	creation refused (enforcement not enforce)	held
C4	ESCALATE: no effect before approval, one after	HTTP 200; effect reached upstream	HTTP 403 $\rightarrow$ approve $\rightarrow$ 200 $\rightarrow$ reuse 403; upstream count 1	held
C5	Unauthorized policy widening rejected	applied (policy v3 loaded)	refused: approval required for 'host.openshell.internal:9'	held
C5b	Host outside signed ceiling rejected	—	refused: outside signed ceiling	held
C6	Provider credential never readable	canary not in sandbox env; attach pending supervisor	attach refused; canary not in env	held
C7	Internet-derived content $\rightarrow$ POST	effect reached upstream	HTTP 403; no effect	held
C8	Native bypass contained	--noproxy POST reached (OCSF: 17); other binary, raw socket, UDP DNS failed	--noproxy POST blocked (OCSF: 17+middleware); other binary, raw socket, UDP DNS failed	held
C9	BULL unavailable $\Rightarrow$ fail closed	—	POST HTTP 403, no effect; create refused; after restart GET 200	held
C10	One correlatable audit trail	8 OCSF L7 events; no authority record	9/9 decisions consistent; 0 effects w/o allow; ledger valid	held

A: OpenShell	as expected	effect happened	as expected	effect happened	effect happened	effect happened	n/a	as expected	effect happened	as expected	n/a	as expected
B: +BULL	held	held	held	held	held	held	held	held	held	held	held	held
	C1	C2	C3	C3b	C4	C5	C5b	C6	C7	C8	C9	C10

Fig. 1. Outcome matrix. Red cells in row A mark effects that OpenShell alone permitted and the composed system blocked; “n/a” marks cases that exist only with BULL.

TABLE III
PHASE B AUDIT JOIN: BULL LEDGER $\times$ OPENSHELL OCSF $\times$ DESTINATION (* = APPROVAL BOUND)

#	Sandbox	Method	Target	BULL	OS L7	OS middleware	Effect	Consistent
1	coder	GET	:40959/guide	ALLOW	ALLOWED	ALLOWED	yes	yes
2	coder	POST	:35911/v1/records	ALLOW	ALLOWED	ALLOWED	yes	yes
3	reader	POST	:35911/v1/records	DENY	ALLOWED	DENIED	no	yes
4	coder	GET	:46731/article	ALLOW	ALLOWED	ALLOWED	yes	yes
5	coder	POST	:35911/v1/records	ESCALATE*	ALLOWED	DENIED	no	yes
6	coder	POST	:35911/v1/records	ESCALATE*	ALLOWED	ALLOWED	yes	yes
7	coder	POST	:35911/v1/records	ESCALATE*	ALLOWED	DENIED	no	yes
8	coder	POST	:35911/v1/x	ESCALATE*	ALLOWED	DENIED	no	yes
9	coder	GET	:40959/after-restart	ALLOW	ALLOWED	ALLOWED	yes	yes

TABLE IV
LATENCY SUMMARY (REQUESTS IN MS; CREATION IN MS)

Metric	Ph.	n	Mean	Med.	p95	p99	Min	Max	σ
GET	A	200	10.23	8.98	16.92	23.38	6.36	31.20	3.49
GET	B	200	17.88	17.07	26.03	30.69	11.43	43.46	4.34
POST	A	200	11.46	9.85	21.19	27.43	6.64	33.86	4.60
POST	B	200	22.65	20.83	34.20	40.64	14.42	65.33	6.60
Create	A	6	914	924	1004	1022	811	1026	67
Create	B	6	986	996	1046	1056	908	1058	48

TABLE V
OVERHEAD (B — A): MEDIAN DIFFERENCE, 95% BOOTSTRAP CI, MANN–WHITNEY U

Metric	Δmed	95% CI	U	z	p	$ r $
GET	+8.08	[+7.12, +8.86]	2525	-15.11	$< 10^{-15}$	0.874
POST	+10.98	[+10.16, +11.72]	2470	-15.16	$< 10^{-15}$	0.877

VIII. RELATED WORK AND POSITIONING

CaMeL separates control and data flow and applies capability policies to agent tool use [18]; the provenance escala-

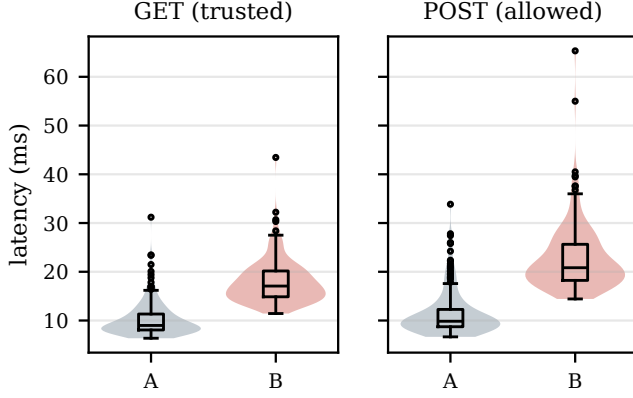


Fig. 2. Per-request latency distributions, $n = 200$ per cell.

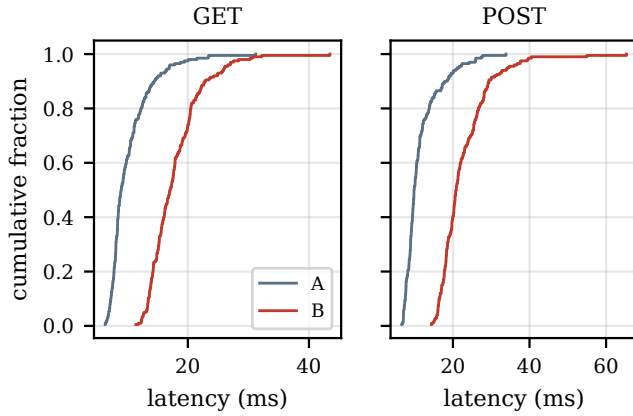


Fig. 3. Empirical CDFs of request latency for Phases A and B.

TABLE VI
LATENCY DRIFT: MEDIAN OF FIRST VS. LAST 50 REQUESTS (MS)

Metric	Phase	First 50	Last 50	Change
GET	A	9.07	9.68	+0.61
GET	B	14.89	17.67	+2.77
POST	A	10.33	9.20	-1.13
POST	B	19.49	25.13	+5.64

tion here is a coarse, sandbox-granular instance of the same idea enforced at the network layer. gVisor and Firecracker isolate untrusted workloads with an application kernel or a lightweight VM [19], [20]; OpenShell composes Landlock, seccomp, network namespaces, and an L7 proxy [1], [21], [22]. The contribution of this report is not a new isolation or authorization mechanism but a tested composition of the two through documented extension points, with destination-judged evidence and a joined audit trail.

IX. LIMITATIONS

Unauthenticated extension transport. The middleware used plaintext gRPC with `allow_insecure_transport`, and OpenShell warns that the service then cannot distinguish

OpenShell from any other network client. A bridge-local process could query BULL verdicts, although it could not alter an enforcement outcome. Production requires HTTPS with gateway-JWT validation; that configuration was not tested.

Scope of mediation. BULL sees HTTP egress and governed control-plane RPCs. It does not authorize process execution inside a sandbox; OpenShell’s Landlock, seccomp, and binary identity contain exec. WebSocket sessions are not mediated. The “exec” half of the architecture’s C7 is therefore OpenShell’s guarantee, not BULL’s.

Weak Phase A evidence for C6. The provider attachment was still awaiting the supervisor when the environment was inspected, so absence of the canary in Phase A does not demonstrate OpenShell’s placeholder mechanism. Phase B’s result rests on BULL refusing the attachment.

Heuristic correlation. OCSF shorthand carries no request identifier; the join uses sandbox, method, target, and time. It was exact here because requests were serialized; under concurrency, identical requests could swap row attribution without changing the counts on which C10 is judged. OCSF JSONL (F5) would remove this ambiguity.

Single host, few runs. One Docker host, no Kubernetes, VM, or hardware-isolated driver. Three complete runs were made; the first surfaced defect 5, and in the last two every case outcome was identical except C10, whose earlier failure was defect 6. Latency figures come from the final run and include shared-host noise. Approval latency (a human decision) is excluded.

X. CONCLUSION

On a live OpenShell v0.1.2 gateway, BULL composed as a narrowing-only layer: every observed effect required both systems’ consent; BULL’s absence failed closed on both data and control planes; provenance-based escalation stopped a post-read exfiltration that OpenShell’s static policy permitted; unapproved widening and credential attachment were refused; and the two systems’ records joined into one consistent trail. The measured cost was a median 8.08–10.98 ms per request in this build, much of it plausibly due to BULL’s linear-time ledger verification. The most consequential risk found was not in the composition but in a default: OpenShell’s audit-mode L7 rules, which BULL now refuses.

DECLARATIONS

Research and authorship. This manuscript describes the BULL project under Luis Minier / Anharmoniclabs. The author is a project stakeholder; the document is not an independent assessment. NVIDIA has not reviewed or endorsed this work; OpenShell is cited as third-party software used under its Apache-2.0 license.

AI assistance. Claude Code (Anthropic) implemented the OpenShell adapters, built OpenShell from source, ran the experiment harness, and drafted this manuscript under the author’s direction. AI-generated text is not experimental evidence; every measurement is generated from the recorded evidence files, whose SHA-256 digests are listed in Supplement S6.

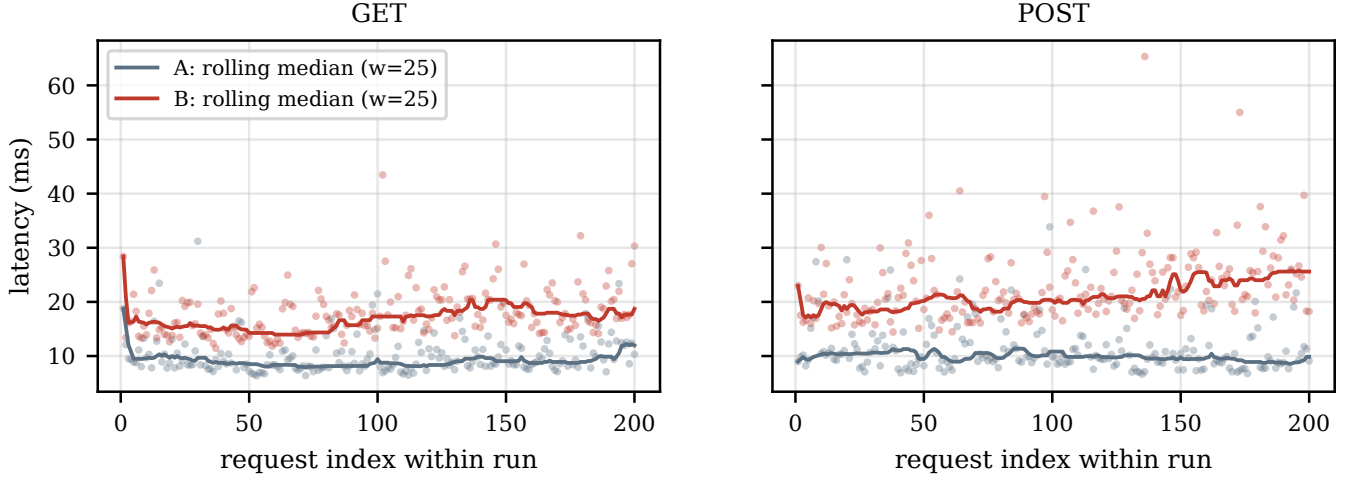


Fig. 4. Latency by request index. Phase B’s rolling median rises through the run while Phase A’s stays flat (Table VI).

TABLE VII
INTEGRATION DEFECTS FOUND AND FIXED

#	Defect	Symptom against the live gateway	Fix
1	Contract capabilities missing from PeerMetadata	Gateway refused to start	Declare <code>openshell.<family>.contract</code> as supported and required
2	Dotted reason codes (<code>bull.denied</code>)	Every request denied as <code>response_reason_code_invalid</code> (closed, but unusable)	Codes renamed <code>bull_*</code> ; grammar test
3	Protobuf-JSON ports are doubles (40605.0)	Endpoint keys would not match later comparisons	Normalize to integers; cover every entry in ports
4	No enforcement-mode check	OpenShell-deny cases silently allowed (F1)	Refuse L7 rules not in <code>enforce</code>
5	Enum form of enforcement not recognized	After fix 4, every sandbox refused	Accept <code>NETWORK_ENFORCEMENT_MODE_ENFORCE</code>
6	Correlator let a denied decision claim a nearby allowed receipt	C10 reported a false “effect without allow”	Assign receipts only to allowed decisions at or after decision time

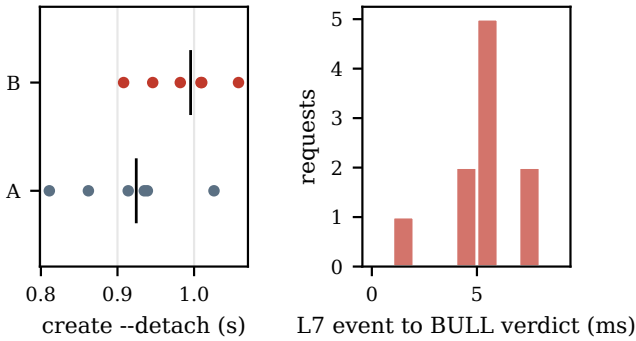


Fig. 5. Left: sandbox admission time (bar = median). Right: interval from OpenShell’s L7 event to BULL’s middleware verdict.

Responsible reporting. No secrets are published. The C6 canary credential was generated per run and verified absent from the evidence set; gateway JWT keys were generated per run and not retained.

APPENDIX A VERSION AND EVIDENCE IDENTITY

A1. BULL. Repository `Anharmoniclabs/BULL`, branch `experiment/bull-openshell-20260928`,

branched from `6682adb`. The final experiment run executed BULL source identical to commit `56faa30` [16].

A2. OpenShell. `github.com/NVIDIA/OpenShell`, tag `v0.1.2`, commit `6648bd0` [1]. Middleware and interceptor stubs were generated from that tag’s `.proto` files.

A3. Evidence. `docs/evidence/openshell-20260928/` [17]; file digests in Supplement S6.

A4. Reproduction.

```
cargo build --release \
  -p openshell-cli -p openshell-gateway \
  -p openshell-supervisor
cargo build --release \
  --target x86_64-unknown-linux-musl \
  -p openshell-sandbox
echo "172.17.0.1 host.openshell.internal" \
  >> /etc/hosts
python tools/openshell_experiment.py \
  --os-src $OS --latency-n 200 \
  --output \
  docs/evidence/openshell-20260928
python \
  tools/build_openshell_ieee_paper.py
```

APPENDIX B CLAIM-TO-EVIDENCE LEDGER

REFERENCES

- [1] NVIDIA, “OpenShell,” tag `v0.1.2`, commit `6648bd0`, 2026. [Online]. Available: <https://github.com/NVIDIA/OpenShell/tree/v0.1.2>

TABLE VIII
CLAIMS, SUPPORTING EVIDENCE, AND BOUNDARIES OF INTERPRETATION

Claim	Evidence	Not established
Effects required both systems' consent	C1–C3, C7; Table III; destination receipts (S3)	Behavior under concurrent or adversarially timed requests.
Escalation admits exactly once	C4 upstream count 1 with reuse refused	Approval UX or human-decision quality.
BULL absence fails closed	C9: POST denied, create refused; orphan event is a denial	Partial failures (slow, not dead, BULL); timeout tuning.
Unapproved widening refused	C5 new port, C5b ceiling, C3b audit mode	Widening through RPCs not bound by the interceptor.
Credential attachment governed	C6-B attach refused, canary absent	OpenShell placeholder mechanism itself (C6-A inconclusive).
Native bypass contained	C8 OCSF evidence, EACCES, blocked UDP	Kernel or container-runtime escape.
Joined audit trail	9/9 consistent, 0 unexplained effects	Exact attribution under concurrency.
Overhead ≈ 8.08 – 10.98 ms/request	Tables IV, V; raw samples S4	Other hosts, authenticated transport, or an $O(1)$ ledger.

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BIOGRAPHY

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SUPPLEMENTARY MATERIAL: COMPLETE DATASETS

All tables below are generated verbatim from the evidence files listed in S6.

S1. Complete Case Results (*cases.csv*)

Ph.	Case	Property	Expected	Observed	Pass
A	setup	create coder	created	ok	yes
A	setup	create reader	created	ok	yes
A	C1	BULL ALLOW + Open-Shell ALLOW executes	reached upstream	http 200; reached=True	yes
A	C2	BULL DENY + Open-Shell ALLOW is blocked	reached (no BULL)	http 200; reached=True	yes
A	C3	BULL ALLOW + Open-Shell DENY is blocked	not reached	http 403; reached=False	yes
A	C3b	L7 rules in default audit mode (OpenShell only)	observe: violation allowed	created=True; admin POST reached=True	yes
A	C4	Escalation without BULL	reached (no approval concept)	http 200; reached=True	yes
A	C7	Internet-derived content - > POST without BULL	reached	reached=True	yes
A	C5	Policy widening without BULL	applied	ok Policy version 3 submitted (hash: de96f00b071a) ok Policy version 3 loaded (active version: 3)	yes
A	C8	Native tool bypass (no proxy, other binary, raw socket, DNS) contained	no unmediated effect; raw socket and DNS fail	{"noproxy_direct": {"rc": 0, "out": "{\n\"stored\": true}"}, "python_socket": {"rc": 1, "out": "raise exceptions[0]\n File \"/usr/lib/python3.12/socket.py\", line 837, in create_connection\n sock.connect(sa)\nPermissionError: [Errno 13] Permission denied"}, "python_http": {"rc": 1, "out": "File \"/usr/lib/python3.12/urllib/request.py\", line 1347, in do_open\n raise URLError(err)\nurllib.error.URLError: <urlopen error [Errno 13] Permission denied>"}, "dns_udp": {"rc": 1, "out": "Traceback (most recent call last):\n File \"<string>\", line 1, in <module>\nOSError: [Errno 89] Destination address required"}, "/v1/x": {"reached_upstream": true, "ocsf_engines": [{"l7": {}}, {"v1/y": {"reached_upstream": false, "ocsf_engines": {}}]}	yes
A	C6	Provider credential not readable (OpenShell placeholder)	secret not in sandbox env	provider_created=True (ok Created provider canary); attach rc=0 coder / canary: persisted (waiting_for_supervisor) Receipt: 35fb08f6-612a-452f-8b36-c2987b4f427d Persisted: 2026-09-28T18:12:43.381635941Z; secret_visible=False	yes
A	C10	Audit trail without BULL (OpenShell OCSF only)	observe: OCSF events, no authority decision record	{"ocsf_http_events": {"l7": 8, "middleware": 0}, "upstream_receipts": 7}	yes
B	setup	create coder	created	ok	yes
B	setup	create reader	created	ok	yes
B	C1	BULL ALLOW + Open-Shell ALLOW executes	reached upstream	http 200; reached=True	yes
B	C2	BULL DENY + Open-Shell ALLOW is blocked	not reached	http 403; reached=False	yes
B	C3	BULL ALLOW + Open-Shell DENY is blocked	not reached	http 403; reached=False	yes
B	C3b	L7 rules in default audit mode are refused at creation	create refused (bull_l7_audit_mode)	created=False; Error: x code: 'The caller does not have permission to execute the specified operation', message: "L7 rules must set enforcement: enforce: ['api:host.openshell.internal', 'docs:host.openshell.internal', 'news:host.openshell.internal']"	yes
B	C4	BULL ESCALATE: no effect before approval; exactly one after	0 before, 1 after approval, 0 on reuse	{"first_http": "403", "pending": 1, "approved": true, "retry_http": "200", "third_http": "403", "upstream_count": 1}	yes
B	C7	Internet-derived content - > POST hits BULL provenance control	escalated	first attempt http 403, reached=False	yes
B	C5	Unauthorized policy widening rejected	rejected pending approval	Error: x code: 'The caller does not have permission to execute the specified operation', message: "approval d029elad44e44632 required to widen policy with ['host.openshell.internal:9']"	yes
B	C5b	Sandbox with host outside BULL ceiling refused	rejected	Error: x code: 'The caller does not have permission to execute the specified operation', message: "policy grants hosts outside the signed BULL ceiling: ['attacker.example']"	yes

Ph.	Case	Property	Expected	Observed	Pass
B	C8	Native tool bypass (no proxy, other binary, raw socket, DNS) contained	no unmediated effect; raw socket and DNS fail	{ "nopproxy_direct": { "rc": 0, "out": "enshell. internal\\", \"layer\\": \"17\\", \"method\\\": \"POST\\", \\\"middleware\\\": \"bull-governance\\\", \"path\\\": \"/ v1/x\\\", \"policy\\\": \"api\\\", \"port\\\": 35911, \"reason_ code\\\": \"bull_approval_required\\\" } }, \"python_socket\": { \"rc\": 1, \"out\": \"raise exceptions[0]\\\\n File \\\\\"usr/ lib/python3.12/socket.py\\\", line 837, in create_ connection\\\\n sock.connect(sa)\\\\nPermissionError: [Errno 13] Permission denied\\\" }, \"python_http\": { \"rc\": 1, \"out\": \"File \\\\\"usr/lib/python3.12/ urllib/request.py\\\", line 1347, in do_open\\\\n raise URLError(err)\\\\nurllib.error.URLError: <urlopen error [Errno 13] Permission denied>\\\" }, \"dns_udp\": { \"rc\": 1, \"out\": \"Traceback (most recent call last):\\\\n File \\\\\"<string>\\\", line 1, in <module>\\\\nOSError: [Errno 89] Destination address required\\\" }, \"/v1/x\": { \"reached_upstream\": false, \"ocsf_engines\": [\"17\", \"middleware\"] }, \"/v1/y\": { \"reached_upstream\": false, \"ocsf_engines\": [] } }	yes
B	C6	Provider attach governed; credential never readable by the agent	attach refused (no credential.use grant); secret not in sandbox	provider_created=True (ok Created provider canary); attach: Error: x provider attachment failed (The caller does not have permission to execute the specified operation); secret_visible=False	yes
B	C9	BULL unavailable - > egress and sandbox creation fail closed	not reached; create refused	post http 403 reached=False; create_while_down=False; after_restart_docs_http=200	yes
B	C10	OpenShell events + BULL decisions + effects correlate	every decision matched, consistent; no unexplained effect; ledger verifies	{ \"bull_decisions\": 9, \"matched_openshell_event\": 9, \"consistent\": 9, \"effects_without_bull_allow\": 0, \"openshell_events_without_bull_decision\": 1, \"orphans_fail_closed\": true, \"ledger_valid\": true, \"ledger_records\": 19, \"ocsf_http_events\": { \"17\": 11, \"middleware\": 10 }, \"upstream_receipts\": 5 }	yes

S2. Audit Join, Phase B (correlation.json)

#	Sbx	Meth.	Resource	Prov.	BULL	Code	Approval	L7	MW	Eff.	Ledger
1	coder	GET	host.openshell. internal:40959/ guide	local_trusted	ALLOW	bull_allowed		ALLOWED	ALLOWED	yes	f0fdd9b52d0c
2	coder	POST	host.openshell. internal:35911/v1/ records	local_trusted	ALLOW	bull_allowed		ALLOWED	ALLOWED	yes	00c66c941d0d
3	reader	POST	host.openshell. internal:35911/v1/ records	local_trusted	DENY	bull_denied		ALLOWED	DENIED	no	bfbdacde8424
4	coder	GET	host.openshell. internal:46731/ article	local_trusted	ALLOW	bull_allowed		ALLOWED	ALLOWED	yes	f50244dca304
5	coder	POST	host.openshell. internal:35911/v1/ records	internet	ESCALATE	bull_approval_ required	36b9cbbcf4819059	ALLOWED	DENIED	no	bb0324f51685
6	coder	POST	host.openshell. internal:35911/v1/ records	internet	ESCALATE	bull_approved	36b9cbbcf4819059	ALLOWED	ALLOWED	yes	804e34a6b6c0
7	coder	POST	host.openshell. internal:35911/v1/ records	internet	ESCALATE	bull_approval_ required	6452be302b3d2906	ALLOWED	DENIED	no	c9ef970549c5
8	coder	POST	host.openshell. internal:35911/v1/x	internet	ESCALATE	bull_approval_ required	fe51d8900f4821ec	ALLOWED	DENIED	no	8f9238c1744b
9	coder	GET	host.openshell. internal:40959/ after-restart	internet	ALLOW	bull_allowed		ALLOWED	ALLOWED	yes	82b54aba3d8d

S3. OpenShell OCSF HTTP Events and Destination Receipts

Phase A events (ocsf-*-A.txt):

Epoch (s)	Sandbox	Engine	Action	Meth.	Target	Reason
1790619153.612	coder	17	ALLOWED	GET	host.openshell.internal:40959/ guide	
1790619153.667	coder	17	ALLOWED	POST	host.openshell.internal:35911/ v1/records	
1790619153.734	reader	17	ALLOWED	POST	host.openshell.internal:35911/ v1/records	

Epoch (s)	Sandbox	Engine	Action	Meth.	Target	Reason
1790619153.793	coder	17	DENIED	POST	host.openshell.internal:35911/admin/delete	L7_REQUEST deny POST host.openshell.internal:35911/admin/delete reason=POST /admin/delete not permitted by policy
1790619154.773	auditor	17	ALLOWED	POST	host.openshell.internal:35911/admin/delete	
1790619154.822	coder	17	ALLOWED	GET	host.openshell.internal:46731/article	
1790619154.883	coder	17	ALLOWED	POST	host.openshell.internal:35911/v1/records	
1790619162.980	coder	17	ALLOWED	POST	host.openshell.internal:35911/v1/x	

Phase B events (ocsf-*-B.txt):

Epoch (s)	Sandbox	Engine	Action	Meth.	Target	Reason
1790619184.386	coder	17	ALLOWED	GET	host.openshell.internal:40959/guide	
1790619184.390	coder	middleware	ALLOWED	GET	host.openshell.internal:40959/guide	
1790619184.437	coder	17	ALLOWED	POST	host.openshell.internal:35911/v1/records	
1790619184.442	coder	middleware	ALLOWED	POST	host.openshell.internal:35911/v1/records	
1790619184.504	reader	17	ALLOWED	POST	host.openshell.internal:35911/v1/records	
1790619184.509	reader	middleware	DENIED	POST	host.openshell.internal:35911/v1/records	middleware_denied:bull-governance:bull_denied
1790619184.567	coder	17	DENIED	POST	host.openshell.internal:35911/admin/delete	L7_REQUEST deny POST host.openshell.internal:35911/admin/delete reason=POST /admin/delete not permitted by policy
1790619184.645	coder	17	ALLOWED	GET	host.openshell.internal:46731/article	
1790619184.649	coder	middleware	ALLOWED	GET	host.openshell.internal:46731/article	
1790619184.704	coder	17	ALLOWED	POST	host.openshell.internal:35911/v1/records	
1790619184.709	coder	middleware	DENIED	POST	host.openshell.internal:35911/v1/records	middleware_denied:bull-governance:bull_approval_required
1790619184.785	coder	17	ALLOWED	POST	host.openshell.internal:35911/v1/records	
1790619184.790	coder	middleware	ALLOWED	POST	host.openshell.internal:35911/v1/records	
1790619184.845	coder	17	ALLOWED	POST	host.openshell.internal:35911/v1/records	
1790619184.852	coder	middleware	DENIED	POST	host.openshell.internal:35911/v1/records	middleware_denied:bull-governance:bull_approval_required
1790619184.957	coder	17	ALLOWED	POST	host.openshell.internal:35911/v1/x	
1790619184.962	coder	middleware	DENIED	POST	host.openshell.internal:35911/v1/x	middleware_denied:bull-governance:bull_approval_required
1790619185.512	coder	17	ALLOWED	POST	host.openshell.internal:35911/v1/records	
1790619185.513	coder	middleware	DENIED	POST	host.openshell.internal:35911/v1/records	middleware_failed: external_service_error
1790619185.877	coder	17	ALLOWED	GET	host.openshell.internal:40959/after-restart	
1790619185.884	coder	middleware	ALLOWED	GET	host.openshell.internal:40959/after-restart	

Destination receipts, Phase A (upstream-receipts-A.json):

Epoch (s)	Method	URL
1790619153.613	GET	http://host.openshell.internal:40959/guide
1790619153.668	POST	http://host.openshell.internal:35911/v1/records
1790619153.735	POST	http://host.openshell.internal:35911/v1/records
1790619154.775	POST	http://host.openshell.internal:35911/admin/delete
1790619154.827	GET	http://host.openshell.internal:46731/article
1790619154.884	POST	http://host.openshell.internal:35911/v1/records
1790619162.980	POST	http://host.openshell.internal:35911/v1/x

Destination receipts, Phase B (upstream-receipts-B.json):

Epoch (s)	Method	URL
1790619184.391	GET	http://host.openshell.internal:40959/guide
1790619184.443	POST	http://host.openshell.internal:35911/v1/records
1790619184.650	GET	http://host.openshell.internal:46731/article
1790619184.791	POST	http://host.openshell.internal:35911/v1/records
1790619185.885	GET	http://host.openshell.internal:40959/after-restart

S4. Raw Request Latency, All Samples in Run Order (latency-raw.json, ms)

#	A GET	B GET	A POST	B POST	#	A GET	B GET	A POST	B POST
1	18.761	28.393	8.973	22.982	101	8.071	17.330	9.711	17.383
2	12.082	13.398	9.828	17.536	102	6.943	43.457	8.507	20.299
3	9.509	16.153	10.169	16.234	103	7.253	27.507	8.778	20.679
4	9.405	16.351	8.196	16.635	104	7.924	17.589	9.818	16.515
5	8.891	21.402	9.279	20.731	105	7.589	19.970	8.505	24.811
6	8.779	18.261	15.123	16.020	106	8.349	15.240	8.915	25.735
7	11.115	13.653	17.178	20.223	107	9.731	15.225	8.958	34.712
8	8.049	13.316	27.422	16.974	108	11.791	20.967	11.244	27.810
9	13.269	15.529	20.176	19.716	109	7.049	15.038	11.374	23.515
10	9.684	13.927	10.448	30.055	110	6.681	16.364	10.160	17.915
11	10.325	17.886	10.835	21.414	111	7.226	17.535	13.773	20.919
12	7.824	22.099	10.167	27.099	112	6.468	24.888	13.641	19.563
13	16.403	25.889	10.569	19.259	113	7.792	26.118	25.978	21.430
14	12.752	13.362	11.420	15.195	114	6.842	16.642	12.016	18.614
15	23.452	14.859	9.379	19.972	115	8.497	22.623	9.829	22.709
16	8.921	15.283	9.203	15.515	116	12.504	18.336	7.558	36.769
17	10.346	13.993	10.329	17.309	117	10.670	18.693	10.093	17.720
18	7.776	13.853	10.335	16.000	118	14.685	14.288	9.650	19.729
19	8.739	12.642	10.983	21.327	119	7.340	15.888	9.443	22.476
20	13.088	13.800	27.796	18.089	120	11.336	14.875	9.346	20.542
21	7.811	16.106	9.571	24.082	121	9.917	17.019	12.103	18.229
22	10.529	16.198	14.016	19.712	122	14.253	22.247	9.542	20.192
23	8.137	13.453	12.275	15.761	123	11.214	20.949	10.281	21.025
24	9.140	20.253	11.287	17.626	124	9.616	20.336	8.150	25.451
25	12.208	19.172	7.755	16.371	125	10.614	20.399	8.504	29.375
26	8.378	19.880	8.067	16.049	126	8.090	22.760	13.026	37.554
27	8.335	19.921	7.070	18.806	127	9.512	16.609	10.924	20.262
28	8.393	15.612	8.885	16.676	128	8.325	20.417	18.845	19.743
29	10.935	15.972	10.640	18.229	129	8.863	17.001	15.106	18.652
30	31.196	19.588	14.116	18.603	130	7.458	13.753	10.135	19.755
31	8.365	14.231	11.323	15.776	131	9.051	19.030	7.042	22.145
32	13.624	13.652	12.788	19.900	132	17.005	18.113	7.740	16.958
33	8.235	13.891	8.191	29.976	133	14.545	25.585	7.114	25.112
34	7.552	13.366	13.844	25.406	134	13.103	26.597	7.026	18.379
35	7.609	14.925	11.756	20.492	135	8.218	20.523	6.645	29.107
36	7.126	12.330	20.759	21.516	136	7.807	20.520	7.098	65.327
37	9.323	11.430	25.899	16.308	137	8.961	20.623	9.421	32.676
38	8.066	17.769	11.603	14.422	138	11.121	15.479	12.054	26.971
39	10.471	20.163	11.027	18.334	139	9.871	17.658	9.773	25.423
40	9.009	17.898	7.586	20.783	140	8.438	15.289	10.440	23.896
41	8.558	14.865	6.908	22.535	141	11.215	16.656	8.841	17.635
42	10.346	14.259	14.448	21.211	142	8.091	21.610	9.020	19.197
43	8.698	18.774	7.712	28.948	143	13.692	20.467	8.675	25.504
44	15.492	15.142	7.259	30.882	144	8.121	20.330	10.846	20.549
45	7.458	13.890	8.557	26.767	145	8.693	24.268	21.169	26.329
46	16.496	13.207	7.157	23.907	146	8.494	30.677	22.363	28.185
47	16.190	12.685	8.424	20.139	147	9.828	26.029	10.469	28.169
48	7.881	13.455	9.462	15.118	148	12.116	19.061	9.241	20.882
49	8.497	13.960	9.843	27.652	149	20.079	19.759	7.436	21.861
50	7.175	13.654	8.224	18.619	150	11.130	17.533	10.500	17.524
51	6.823	21.881	12.149	20.690	151	8.513	16.252	7.457	18.400
52	6.583	22.633	16.906	35.993	152	9.151	16.823	11.326	22.890
53	6.359	15.158	15.633	28.010	153	8.381	15.994	9.706	26.394
54	6.966	15.884	13.855	22.119	154	8.388	18.002	13.209	28.886
55	6.897	15.306	10.556	21.346	155	11.850	17.657	11.032	28.153
56	6.732	13.706	7.740	14.830	156	11.451	17.459	7.287	29.185
57	8.383	12.483	9.161	16.623	157	7.988	26.985	9.482	26.337
58	10.230	12.759	7.053	17.064	158	7.639	20.177	7.413	19.426
59	8.336	11.845	8.959	17.768	159	7.564	19.360	7.582	18.346
60	8.382	12.293	8.005	17.006	160	8.471	15.288	19.453	21.920
61	8.036	13.138	7.365	14.476	161	8.723	17.825	11.950	20.278
62	7.495	12.373	9.501	18.765	162	8.909	16.225	18.730	24.008
63	10.908	18.720	15.446	21.850	163	10.005	14.169	8.570	19.676
64	11.118	19.909	24.221	40.497	164	9.490	14.315	9.196	32.797
65	9.298	24.941	13.106	18.139	165	11.326	14.366	8.901	26.548
66	9.606	19.579	12.276	19.980	166	12.871	22.275	8.736	24.332
67	10.025	19.171	17.220	16.153	167	10.502	21.282	9.533	23.036
68	7.708	14.775	12.482	16.229	168	8.095	23.506	9.894	25.064
69	7.405	13.444	19.582	16.137	169	7.385	20.225	7.770	26.089
70	7.681	14.092	17.862	18.762	170	7.116	19.980	7.092	19.509
71	8.181	14.331	8.999	16.922	171	8.575	16.198	10.458	18.261

#	A GET	B GET	A POST	B POST	#	A GET	B GET	A POST	B POST
72	6.396	14.197	10.379	16.094	172	9.085	17.118	7.042	34.170
73	8.128	14.365	9.689	21.216	173	13.911	17.859	7.297	54.996
74	8.061	13.667	10.118	22.156	174	8.132	14.874	7.630	20.438
75	7.120	13.519	9.125	27.987	175	13.108	15.021	7.679	21.329
76	11.115	17.350	9.606	28.316	176	8.413	16.825	6.765	21.003
77	7.356	22.122	8.946	18.191	177	9.282	15.678	13.433	25.784
78	7.745	20.543	9.429	23.746	178	7.923	23.702	13.665	25.384
79	7.944	19.535	7.856	20.253	179	13.546	32.220	20.082	17.963
80	13.767	17.324	16.861	20.557	180	11.853	21.489	15.195	25.601
81	7.838	16.620	11.333	21.830	181	17.836	20.372	7.894	37.606
82	16.916	15.242	14.477	19.825	182	16.940	16.471	7.681	29.374
83	15.090	15.592	10.820	17.941	183	7.790	16.152	7.764	33.889
84	10.542	17.801	22.191	27.223	184	10.549	14.339	9.848	23.167
85	8.770	17.586	12.048	20.390	185	9.878	15.723	9.077	26.266
86	10.895	14.006	11.372	21.557	186	15.538	14.138	17.198	28.500
87	8.112	12.416	9.138	20.557	187	12.587	19.041	7.772	20.875
88	7.309	15.695	8.885	16.130	188	9.299	18.724	8.839	28.133
89	8.175	18.425	17.580	17.707	189	7.095	22.936	8.567	31.442
90	10.809	17.302	9.822	17.040	190	12.226	25.781	8.766	32.215
91	8.637	21.284	9.967	16.369	191	14.394	20.135	10.899	21.311
92	14.036	18.601	8.776	17.771	192	9.380	20.444	7.662	23.127
93	7.269	15.815	9.631	18.958	193	12.771	16.938	25.801	26.049
94	8.542	17.898	13.989	18.165	194	23.381	21.944	19.047	20.329
95	8.114	15.616	18.521	17.790	195	12.467	16.960	21.662	25.188
96	9.433	15.634	14.727	21.731	196	11.961	17.676	9.211	26.643
97	10.372	14.632	10.972	39.468	197	12.553	17.142	10.449	24.594
98	19.423	17.883	8.511	29.163	198	8.573	17.640	11.746	39.690
99	14.908	13.316	33.856	22.711	199	12.377	27.050	11.336	18.292
100	21.481	15.140	9.845	18.540	200	10.308	30.317	9.099	18.214

S5. Sandbox Creation Samples (s) and L7-to-Verdict Intervals (ms)

#	A	B
1	1.026	1.010
2	0.935	0.908
3	0.939	1.009
4	0.914	0.946
5	0.862	1.058
6	0.811	0.982

L7-to-verdict intervals, Phase B, in event order: 4, 5, 4, 5, 5, 7, 5, 1, 7, 5

S6. Evidence File Integrity (SHA-256)

Every number in this paper was computed from these files by `tools/build_openshell_ieee_paper.py`.

File	Bytes	SHA-256
README.md	1396	d4e8adae9887f56edb225a7d8114171c71bbc66671dfc924fe7f3bad65390218
bull-audit.jsonl	312291	340a7cc304af900367c1ea221979cb81eab7f7808d4009f0ac067560669bd242
bull.log	52745	8a1cbc68fd43d3cb92a416856a0f6b5fe1e9b4347402bb32ef383b0ed3bdeda7
cases.csv	6404	23c4de9120031d4943246a07991e2f76b80a23c654049508d7df6d69101ff3d3
correlation.json	6496	c718ac4013c8a417fad372a432c0ee43c2df65e53989910653c1c1a608a4d666
gateway-A.log	242308	9e8d586e4db5ea8a3347e92b487c0454162f2e92f76261c0a214a925b7659a0b
gateway-A.toml	496	6ffdc55cfc57e629cd0a6e64f36115b2a15866d949e34ad9d94f6788b6d88f3c
gateway-B.log	261142	93b8de8ab71267cd32867087d7ee2430d1e5957a09ad3a78587224ce4b02621d
gateway-B.toml	1628	fcc36360b7265a4c24b77f5def645cb51e3822774da0f68e3ef020ad96d79398
latency-raw.json	6662	f3c7d17932a1c567d9ad2003c1b356cf0917cfc0a8ddcd045746094fdd77f56c
latency.csv	503	8c20b41183132722be2dbdb08bd0e840bb31392b616767ae200653d5ea04dac5
ocsf-auditor-A.txt	3463	4aa278f9af100986efa226cc29f019e7b8ed119f2d27758ee4b8010e0e371d22
ocsf-auditor-B.txt	0	e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
ocsf-coder-A.txt	11799	69ac92422596e1c0123b11929a4bd077d405695cd2e3aff847b5185820703d43
ocsf-coder-B.txt	15885	990e98f0521ebfd1a4d30b4ff2791644d9e20d4bf6cd53461061da92ff194e1f
ocsf-reader-A.txt	3458	48e6de30cece45547ca46bddd46f072db52d3ac4ab2bc3713c4f9f847c7a6e6
ocsf-reader-B.txt	3148	8f6c8810952f285096b751e1c44edd2009f1653663805a81ab9c9835c362d35b
policy.yaml	977	c77976ada5d5855020934678b4f8e05f2061301ef82851c5130e3bdaf7053c7
results.json	17122	51e0f0c49c85cb296f93af6abdd4836b15c2bb54ce37d4485e24964531c81358
upstream-receipts-A.json	798	51db4130544dfb67d84530bbfb4f177bc89e8451f1f18cf316d9f1e88961648f
upstream-receipts-B.json	574	fdbe5c9ef8862eda1fae1093302b6aafabbe135334a5170eb853ff4f431158df